

ABINASH BEHERA

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EDUCATION

Indian Institute of Information Technology, Nagpur	Aug 2023 – May 2027
Bachelor of Technology in Computer Science and Engineering - CGPA 9.00	
D.A.V Public School, Kolkata	2021
Senior Secondary Examination - 95%	
D.A.V Public School, Kolkata	2019
Secondary Examination - 97%	

RELEVANT COURSEWORK

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|--------------------|---------------------------|------------------------|----------------------------|
| • DSA | • DBMS | • Operating Systems | • Computer Networks |
| • Machine Learning | • Artificial Intelligence | • Discrete Mathematics | • Probability & Statistics |

PROJECTS

AI Data Analyst Agent 🤖 | Python, LangGraph, LLaMA 3, Pandas, Plotly, Streamlit May 2026 – Present

- **Orchestrated** an autonomous data pipeline using **LangGraph** and **LLaMA 3**, integrating a multi-agent system (Planner, Analysis, Visualization, Insight) for automated **Exploratory Data Analysis (EDA)**.
- **Executed** tool-augmented LLM reasoning to eliminate hallucination, leveraging **Pandas** for deterministic statistical computations while delegating natural language interpretation to the **LLM**.
- **Constructed** a memory-aware Q&A interface via **Streamlit**, enabling natural language data queries while dynamically rendering context-aware **Plotly** charts and automated cleaning recommendations.

AutoMedRAG 🤖 | FastAPI, FAISS, BM25, Cross-Encoder, NVIDIA LLMs, Streamlit Jan 2026 – Present

- **Architected** a medical AI system over **35M+ PubMed articles** using hybrid retrieval (**FAISS** dense + **BM25** sparse + **cross-encoder** reranking) across **20+ conditions**, achieving **3× higher relevance accuracy**.
- **Designed** a **4-service microservices architecture** with fallback mechanisms for **>99% uptime**, incorporating multi-turn conversational context management.
- **Developed** a production-grade **FastAPI backend** integrating **NVIDIA LLMs (5–6s latency)** and a **Streamlit frontend** supporting **3 search modes**, API health checks, and text-to-speech.

Real-Time Sign Language Translator 🤖 | Python, OpenCV, MediaPipe, TensorFlow, LSTM, Streamlit May 2025 – Oct 2025

- **Engineered** a real-time **ASL recognition system** using **Python**, **OpenCV**, and **MediaPipe**, extracting **1,662 body landmarks per frame** from **8,300+ training samples**.
- **Trained** a stacked **LSTM deep learning model** with **TensorFlow/Keras**, applying dropout regularization, learning rate scheduling, and data augmentation to achieve **97% test accuracy**.
- **Devised** a modular **computer vision pipeline** with **real-time inference at 25+ FPS**, featuring prediction smoothing, confidence scoring, and a **Streamlit** web interface.

Money Laundering Transaction Detector 🤖 | PyTorch, PyTorch Geometric, Streamlit Jan 2026 – Feb 2026

- **Produced** a hybrid **GraphSAGE + Transformer ensemble** to detect illicit Bitcoin transactions on the **Elliptic dataset** (203k nodes, 234k edges), modeling transaction flows as a directed temporal graph.
- **Assembled** a **3-layer GraphSAGE encoder** and **3-layer Transformer** with learned temporal encodings, achieving **F1 of 0.91** and **93.4% recall**, outperforming **GCN baseline by 88.3%**.
- **Deployed** an interactive **Streamlit dashboard** with real-time transaction network visualization, suspicion scoring, and adjustable detection threshold for live fraud flagging.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript
Web Development: React.js, HTML5, CSS3, Node.js, Express.js, REST APIs, MongoDB
Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, FAISS, Feature Engineering
Generative AI: LangChain, LangGraph, RAG Pipelines, FastAPI, OpenAI API, Hugging Face
Tools: Git, GitHub, Flask, Streamlit

ACHIEVEMENTS & CERTIFICATIONS

- **Certifications:**
 - Machine Learning — *Coursera (Andrew Ng)*
 - Generative AI Fundamentals — *Udemy*
 - NVIDIA Fundamentals of Deep Learning — *NVIDIA*
 - Fundamentals of Accelerated Computing with CUDA — *NVIDIA*
- Participated in the **Amazon ML Challenge 2025**, developing ML models for large-scale real-world datasets and achieving a SMAPE score of 51 through advanced feature engineering and model tuning.